

Exam 2

ELEE 4210/6210: Linear Systems

Date: Sept. 19, 2025

Time allowed: 50 minutes

Note: Exam is closed book & closed notes. You are only allowed to use a calculator and ONE formula sheet.

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Problem 1 [2+1.5+1.5=5 points]: The response of a causal LTI system to a step input, i.e., $x(t) = u(t)$, is given by $y(t)$ shown below. For this system:

- (i) Sketch the impulse response of the system, i.e., $h(t)$.
- (ii) Determine whether or not the system is BIBO stable. Why?
- (iii) Determine whether or not the system is causal. Why?

$$x(t) = u(t)$$

$$y(t) = 2tu(t) - 4(t-2)u(t-2) + 2(t-4)u(t-4)$$

$$h(t) = \frac{d}{dt} y(t)$$

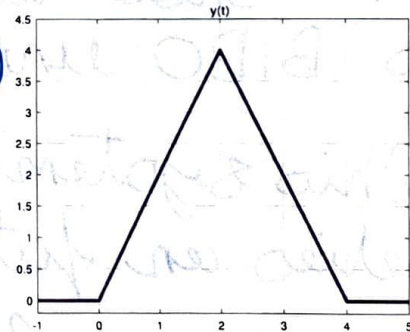
$$= \frac{d}{dt} (2tu(t)) - \frac{d}{dt} (4(t-2)u(t-2))$$

$$+ \frac{d}{dt} (2(t-4)u(t-4))$$

$$= 2u(t) + 2t\delta(t) - \frac{d}{dt} ((4t-8)u(t-2))$$

$$+ \frac{d}{dt} (2t-8)u(t-4) = 2u(t) - \frac{d}{dt} (4tu(t-2))$$

$$+ \frac{d}{dt} (8u(t-2)) + \frac{d}{dt} (2tu(t-4)) - \frac{d}{dt} (8u(t-4))$$



$$= 2u(t) - 4u(t-2) - 4t\delta(t-2) + 8\delta(t-2) + 2u(t-4) + 2t\delta(t-4) - 8\delta(t-4)$$

$$h(t) = 2u(t) - 4u(t-2) + 8\delta(t-2) + 2u(t-4) - 8\delta(t-4)$$

$$h(t) = 2u(t) - 4u(t-2) + 2u(t-4) + \delta(t-2) - \delta(t-4)$$

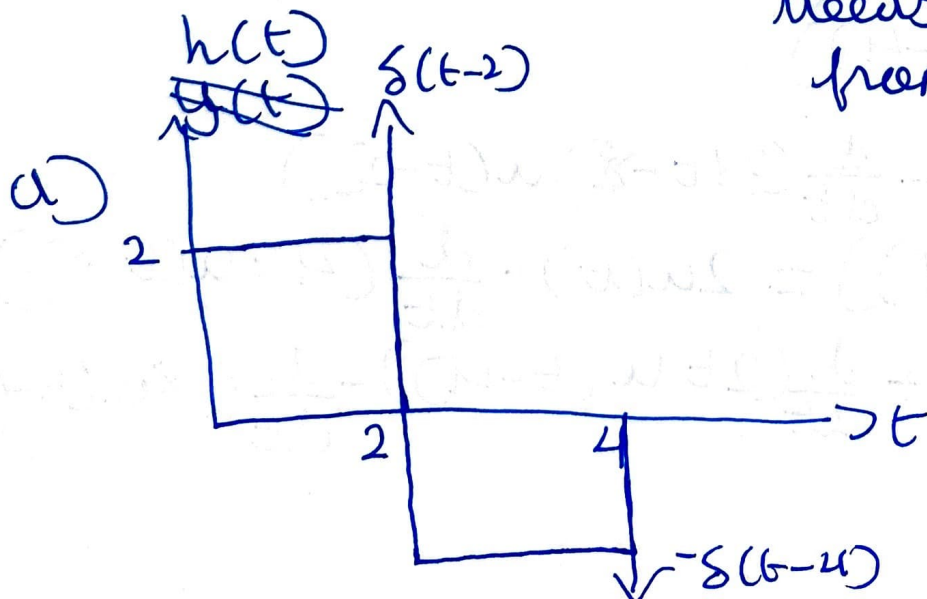
b) To check BIBO stability,

$$\int_{-\infty}^{\infty} |h(t)| dt = \int_{-\infty}^{\infty} [2u(t) - 4u(t-2) + 8\delta(t-2) + 2u(t-4) - 8\delta(t-4)] dt$$

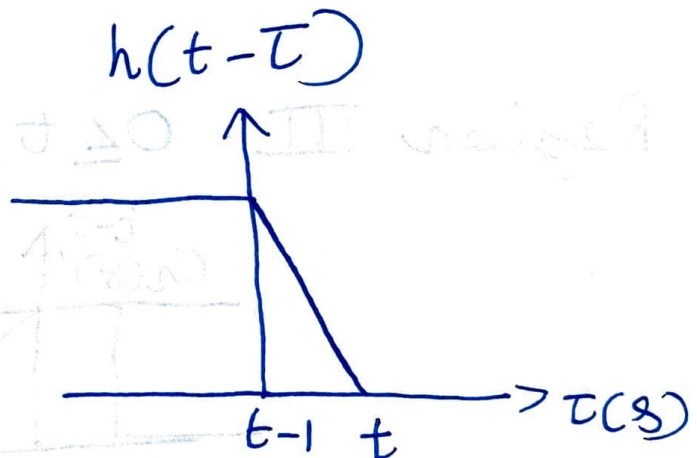
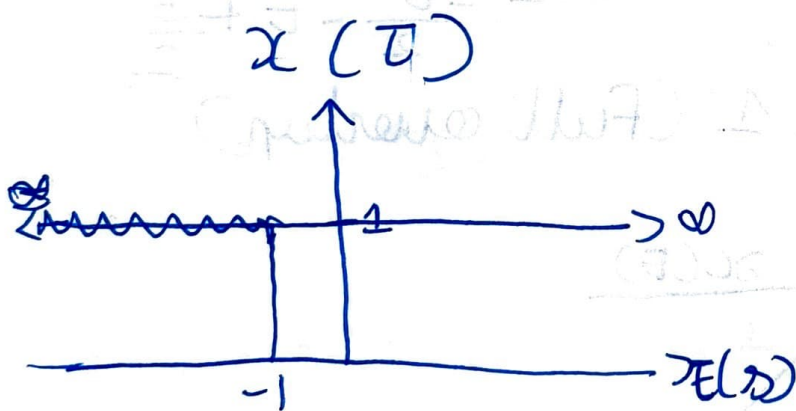
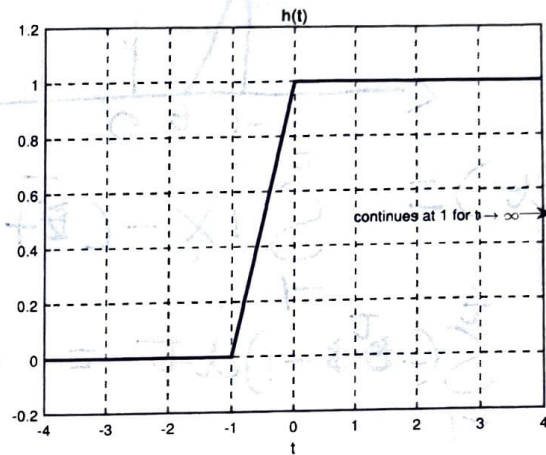
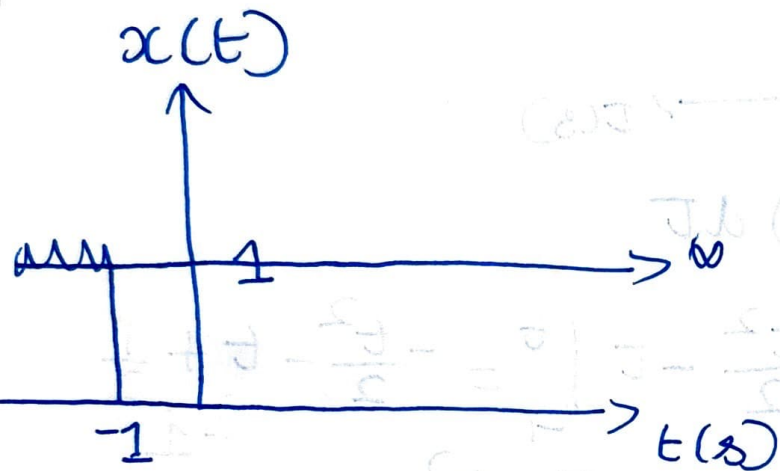
As any of these impulse or step functions can shoot up to infinity, this system is BIBO unstable.

c) This system is non causal as it relies on future values

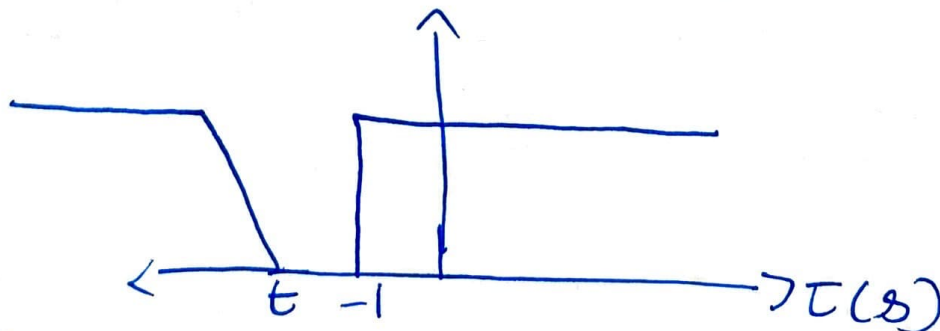
eg: from $h(t) \rightarrow -4u(t-2) \dots$
Needs value $\uparrow$
from $t=2$



Problem 2 [5 points]: For $h(t)$ shown below and $x(t) = u(t+1)$ (that is a delayed step function), use the graphical convolution method to find and plot $y(t) = x(t) * h(t)$. Show all the details of your work.

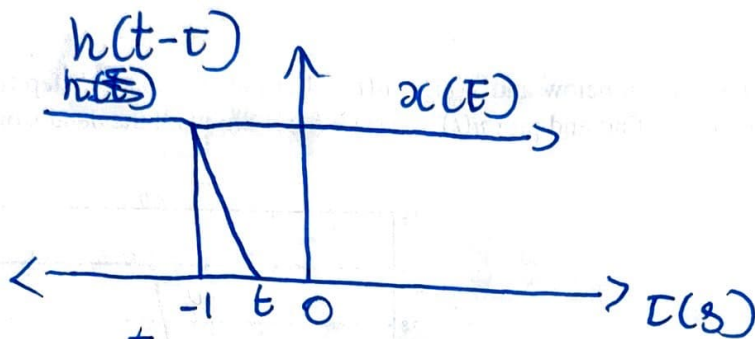


$h(t) = r(t+1) = (t+1)u(t+1)$
 Region I $t < -1$ (No overlap)



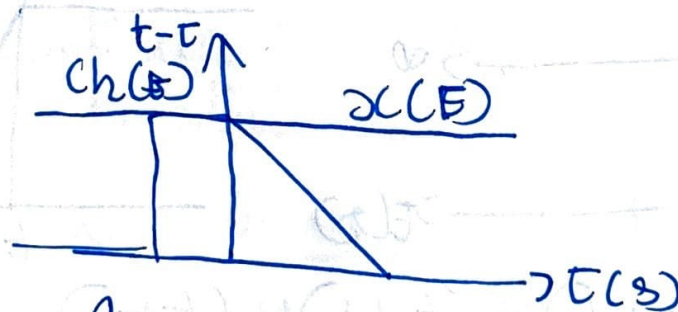
$$y(t) = 0$$

Region II $-1 \leq t \leq 0$ (Partial Overlap)



$$\begin{aligned} \therefore y(t) &= \int_{-1}^t 1 \times -(\tau + 1) d\tau \\ &= \int_{-1}^t (-\tau - 1) d\tau = -\frac{\tau^2}{2} - \tau \Big|_{-1}^t = -\frac{t^2}{2} - t + \frac{1}{2} \\ &= -\frac{t^2}{2} - t + \frac{3}{2} \end{aligned}$$

Region III $0 \leq t \leq 1$ (Full overlap)

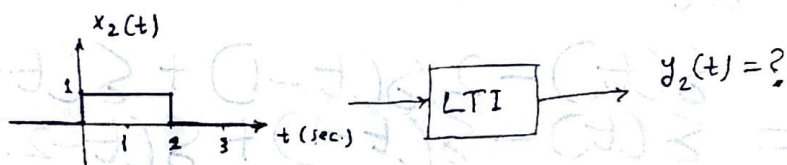
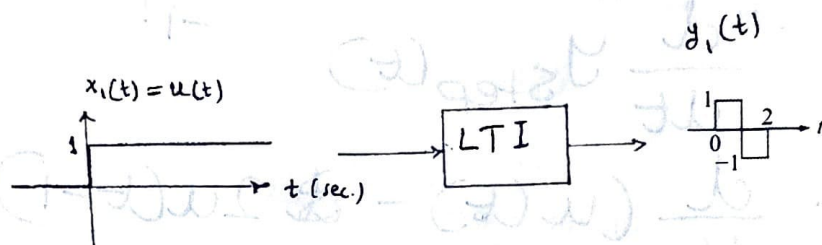


$$y(t) = \int_0^1 1 \times 1 d\tau = \int_0^1 d\tau = \tau \Big|_0^1 = 1$$

Problem 3 [3+2=5 points]: You observe an unknown LTI system and notice that the response to the input x_1 is the output y_1 as shown below.

(a) If you apply another input $x_2(t)$, as shown below, to the same LTI system, determine and sketch the new output response $y_2(t)$.

(b) Determine and sketch the impulse response of this system, i.e., $h(t)$.



$$h(t) = \frac{d}{dt} y_1(t) = \begin{cases} 1 & 0 \leq t \leq 1 \\ -1 & 1 \leq t \leq 2 \\ 0 & \text{otherwise} \end{cases}$$

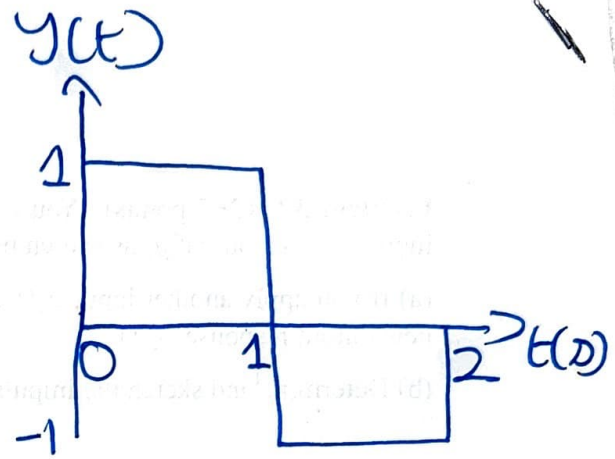
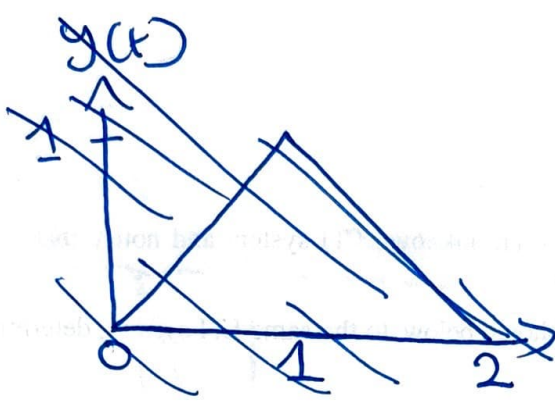
$$x_2(t) = \begin{cases} 1 & 0 \leq t \leq 2 \\ 0 & \text{otherwise} \end{cases}$$

$$\therefore y(t) = \int_0^1 x(\tau) h(t-\tau) d\tau$$

$$+ \int_1^2 x(\tau) h(t-\tau) d\tau = \int_0^1 (1 \times 1) d\tau + \int_1^2 (1 \times -1) d\tau$$

$$= \tau \Big|_0^1 + (-\tau) \Big|_1^2 = 1 - 1 = 0$$

a)



b) $h(t) = \frac{d}{dt} y_{\text{step}}(t)$

$$h(t) = \frac{d}{dt} (u(t) - 2u(t-1) + u(t-2))$$

$$h(t) = \delta(t) - 2\delta(t-1) + \delta(t-2)$$

$$h(t) = \delta(t) - \delta(t-1) + \delta(t-2)$$

